Analyzing Traffic Incidents in Izmir

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***Abstract*—The abstract goes here.**
Our code for this project is available at [1].

I. Introduction

Traffic incidents are a prevalent problem in Izmir, imposing substantial costs on both citizens and the municipality, including life-threatening risks, economic losses, and operational burdens. Providing concise, data-driven evidence can accelerate decision-making and lead to precautionary actions that lower incident rates. While national studies exist, recent work centered on Izmir's incident patterns is scarce. This study analyzes a dataset of traffic incidents from the Izmir metropolitan area.

In this study, the incident records from the Izmir Municipality Open Data Portal are analyzed, including timestamps, location data and incident information. Our goal is to determine traffic incident patterns, summarizing locations and times at which risk concentrates across the metropolitan area.

Fig. 1. Map of the Izmir metropolitan area.

II. Background and Related Work

Numerous studies have investigated the spatial and temporal relationships of traffic incident occurrence using various analytical and data-driven methods.

In a study conducted by Cakıcı on Izmir [1], it is shown that the highest number of accidents in the years between 2020-2023 occurred in the autumn and summer seasons. In addition, least number of traffic accidents occurred during 2020 spring season due to the COVID-19 pandemic.

Another study conducted by Haybat and Karakas [2] reveals that incidents are focused on the high mobility regions of Izmir, especially in the south of Karsıyaka district, all over district of Konak, northeast of Karabaglar district, and northwest of Buca. Furthermore, the study reveals that while the number of incidents increased between 2010 and 2014, the total coverage area declined, suggesting a higher spatial density of incidents in later years.

According to a study performed by Kumar and Toshniwal [3] on road accidents in Gujarat, the time periods between 20:00 to 22:00 and 22:00 to 00:00 were identified as the most accident-prone invervals when the 24-hour day was divided into two-hour periods.

Cardelino (1998) [4] analyzes hourly traffic-counter data and reports the familiar weekday rush-hour double peak, contrasted with a relatively flatter weekend profile. Although framed around emissions, the study highlights the value of distinguishing traffic pattern types in time-of-day analysis.

Retallack and Ostendorf [5] report that crash frequency increases non-linearly as congestion rises. Their models, estimated on over five million hourly observations from 120 intersections, show that higher congestion is associated with disproportionately larger gains in expected crash counts, especially at the most congested levels. In short, the results indicate a positive, non-linear correlation between traffic congestion and crash rates.

III. Data Set Description

The data set used in this study was provided by the Open Data Portal of Izmir Metropolitan Municipality and authored by Izmir Transportation Center Directorate (IZUM) [6]. The data set includes detailed records of vehicle breakdowns and traffic accidents that occurred within the metropolitan area of Izmir. The metropolitan area of Izmir comprises the central districts of Çiğli, Karşıyaka, Bayraklı, Bornova, Konak, Buca, Karabağlar, Gaziemir, Balçova and Narlıdere, as illustrated in Figure 1.

At the time of our access to the data set, October 11, 2025, the observations cover the period between December 1, 2021,

[1]https://github.com/HuseyinYontar/Analyzing_Traffic_Incidents_in_Izmir

TABLE I
DESCRIPTION OF EACH FEATURE IN THE RAW DATA SET.

Attribute	Type	Median	Mode	Number of Distinct Values	Number of Missing Values	Minimum	Maximum
Date	Interval	2023-07-18	2022-04-29	1397	0	2021-12-01	2025-09-28
Street	Nominal	-	Yeşildere Caddesi	183	0	-	-
Direction	Nominal	-	Merkez	157	16	-	-
Location	Nominal	-	Hilal Köprü Üstü	2032	14	-	-
Incident Type	Nominal	-	Maddi Hasarlı	30	0	-	-
Incident Time	Interval	15:30	17:29	1344	0	00:00	23:59
Intervention Time	Interval	15:52	18:10	1295	5193	00:00	23:59

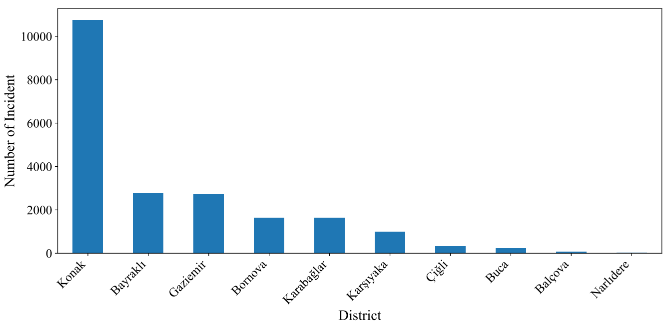

Fig. 2. Number of incidents per district between 2021-12-01 and 2025-09-28.

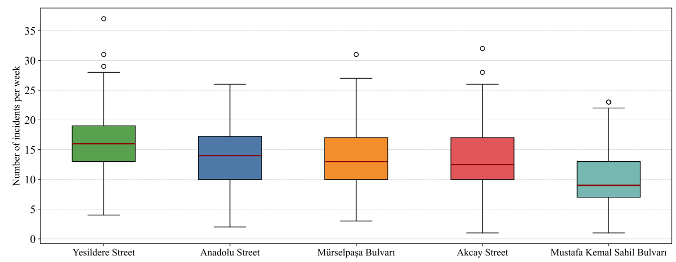

Fig. 3. Box-plot showing weekly incident counts for top 5 most incident-dense streets in İzmir

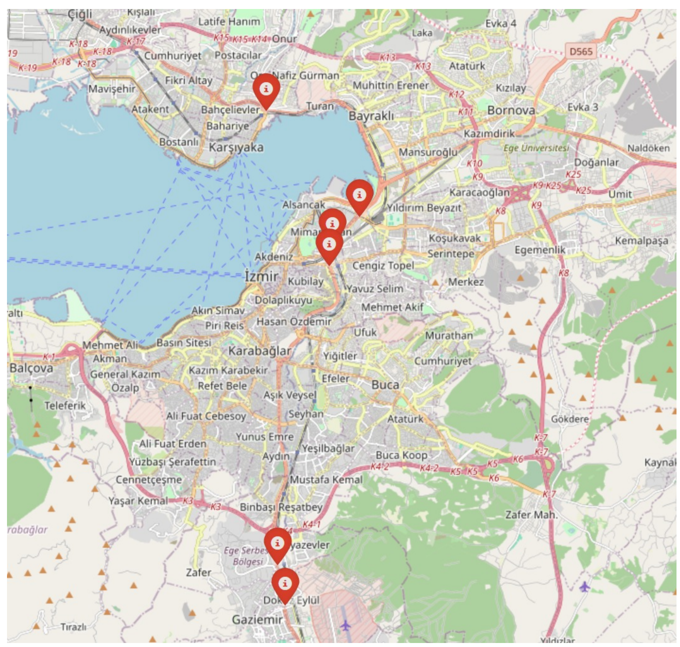

Fig. 4. Demonstrating Hotspot Incident Points of Izmir

and September 28, 2025. The raw data set consists of 21161 samples and 7 attributes. Each sample contains the street name, the direction of travel with 16 missing values, the location description with 14 missing values, the type of incident as nominal data type and the date of the incident, the accident time, and the intervention time with 5193 missing values as interval data type.

The descriptive information regarding the data set including mode and median values can be found in tabular form in Table I.

IV. PREPROCESSING

As part of the preprocessing stage, the dataset was examined to understand spatiotemporal incident patterns across Izmir's metropolitan districts and its most incident-dense streets.

Figure 2 demonstrates the total number of incidents by metropolitan districts. The bar chart shows that the highest number of incidents occurred in Konak, followed by Bayraklı and Gaziemir. Bornova, Karabağlar, and Karşıyaka also contribute considerably to the incident counts. In contrast, Çiğli, Buca, and Balçova contribute only a small share. The bar chart reveals that the total number of incidents in Konak is higher than the combined total of all other districts, indicating that Konak is the most incident-dense district in the Izmir metropolitan area.

Figure 3 illustrates the box plot of the weekly incident counts for the five most incident-dense streets of Izmir. From the box plot, Yesildere Street has the highest median weekly incident count, indicating a higher typical number of incidents than the other streets. Additionally, both Akçay Street and Mustafa Kemal Sahil Boulevard display lower minimum weekly incident counts compared to the other streets. Another notable pattern is that outliers are observed only on the upper end of the distributions, with no lower-end outliers present, indicating a right-skewed pattern in weekly incident counts.

Figure 4 displays the most incident-dense locations in Izmir: Mürselpaşa Boulevard on Hilal Bridge overpass with 954 incidents, Yeşildere Avenue on Tepecik Bridge overpass with 949 incidents, Mürselpaşa Boulevard on DGM Bridge overpass with 886 incidents, Akçay Avenue inside the Free Zone underpass with 521 incidents, Akçay Avenue inside the Gaziemir underpass with 497 incidents, and Anadolu Avenue after Naldöken Bridge with 484 incidents. This figure is crucial

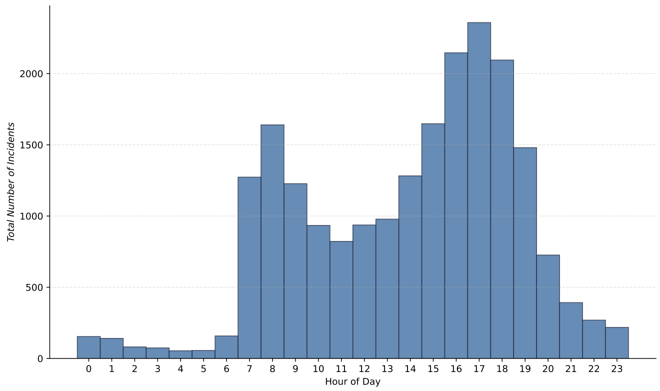

Fig. 5. Hourly incidents histogram

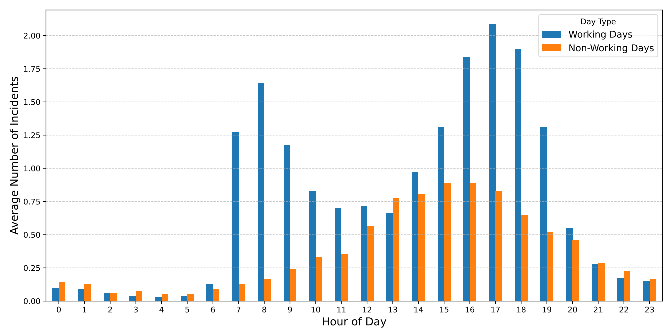

Fig. 6. Hourly incidents bar chart comparing working days and non-working days.

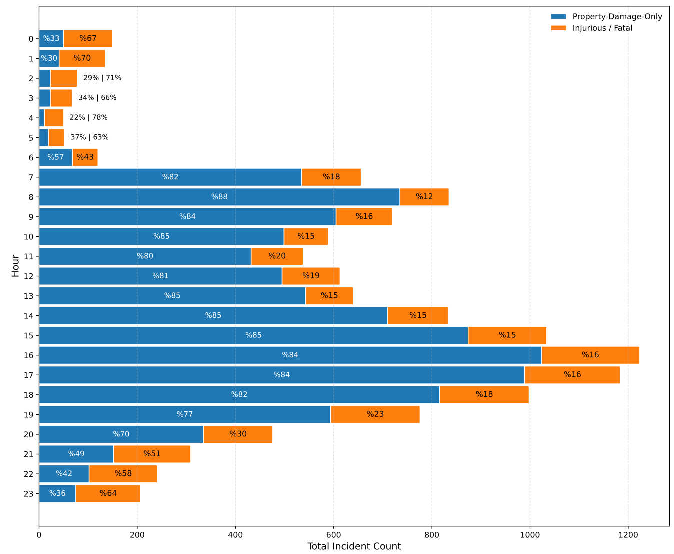

Fig. 7. Distribution of total accidents per hour including their accident type percentages

for pinpointing hotspots, where focused interventions produce the greatest impact per unit of effort and cost.

Figure 5 illustrates the distribution of incident counts across the hours of the day. The number of incidents exhibits an upward trend during the early morning hours, reaching its first peak at around 08:00. After this point, incident occurrences decline until midday. Following noon, incident counts rise again, with the highest rates observed around 17:00, which aligns with the end of typical working hours. Incident counts then decline sharply throughout the evening and continue to decrease into the late-night period.

Figure 6 compares the average hourly incident counts on working and non-working days. On working days, incidents begin to increase after 07:00, reach a morning high at 08:00, and then decline toward 11:00. Thereafter, average incident count gradually rise again and reaches its peak at 17:00. Following this point, incidents decline sharply until around 20:00 and continue decreasing into the late hours.

On non-working days, including weekends and national holidays, total incident numbers rise steadily through the morning, peak around 16:00, and then gradually decline into the late-night hours. These findings on İzmir's traffic patterns, particularly the pronounced mid-afternoon peak on non-working days, are consistent with Cardelino's weekend traffic volume distribution results [4].

By examining hourly incident patterns, weekdays exhibit a double peak that aligns with established weekday congestion signatures, while weekends show a flatter profile with a pronounced mid-afternoon high [4]. Retallack and Ostendorf [5] further show that expected crash counts rise non-linearly as congestion intensifies. According to their analysis, heavy traffic yields larger marginal increases in expected crashes, indicating a positive, non-linear congestion and crash rates.

Figure 7 shows how the distribution of accident types varies by hour of day, using a stacked bar chart of hourly totals with within-hour percentage shares. In this chart, injury and fatal incidents are combined into a single injury/fatal category, while property-damage-only crashes remain as a separate group due to their high frequency. Less common incident types such as pile-up and similar categories are excluded from the figure, but this omission is not expected to affect the overall pattern due to their low occurrence. Although the number of accidents per hour decreases after 22:00, the analysis indicates that the ratio of injury or fatal crashes to property-damage-only incidents increases until 06:00. This figure suggests that while daytime traffic density contributes to high amount of incidents, nighttime incidents are observed to have smaller amount of incidents with higher injury and fatal share.

Figure 8 presents two bar charts: the top panel shows the number of distinct streets by district, and the bottom panel shows the average number of incidents per street. The first inference concerns Konak, across both panels, Konak ranks first in both street count and average incidents per street. Because overall incident totals are influenced by both the number of streets and the per-street average, Konak's leading total in Figure 2 is consistent with these results.

Another inference concerns Bayraklı. Figure 2 shows Bayraklı has the second-highest total incidents. In Figure 8, Bayraklı ranks sixth in street count and second in average incidents per street. This combination indicates very high

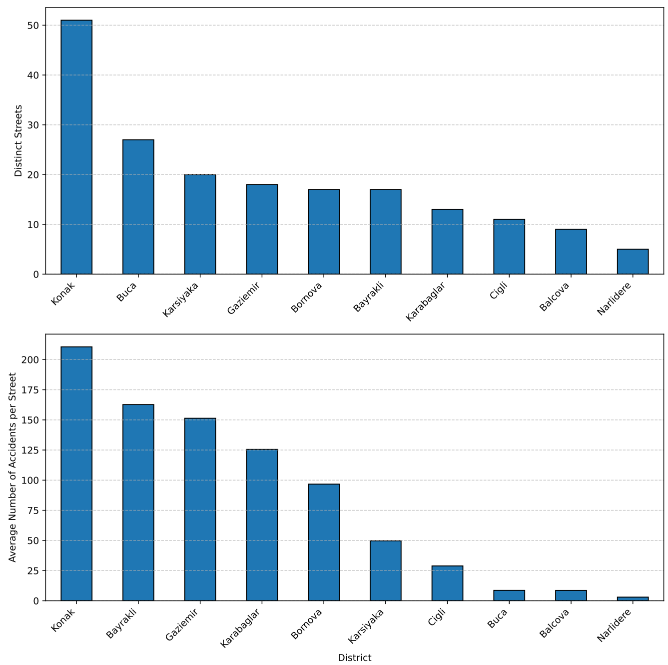

Fig. 8. Number of streets (top) and average incidents per street (bottom), by district

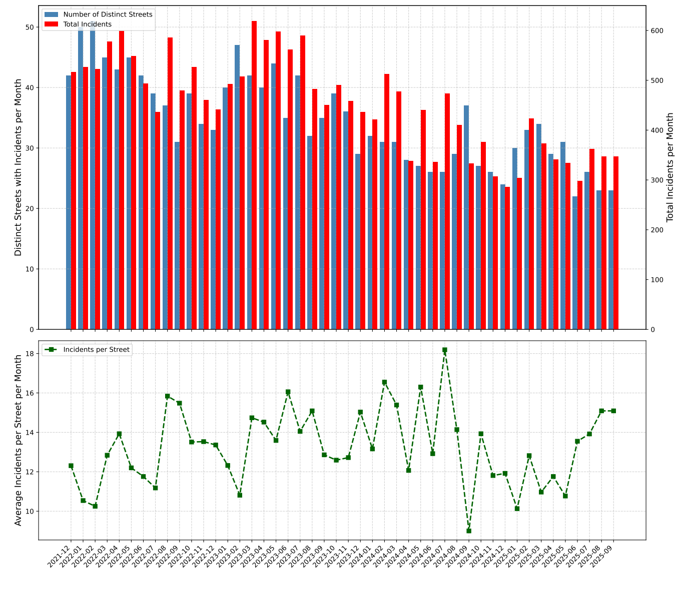

Fig. 9. Monthly incident summary for İzmir, showing total incidents and the number of distinct streets with incidents (at top, combined together), and the average number of incidents per street (at bottom), computed as total incidents divided by distinct incident-involved streets.

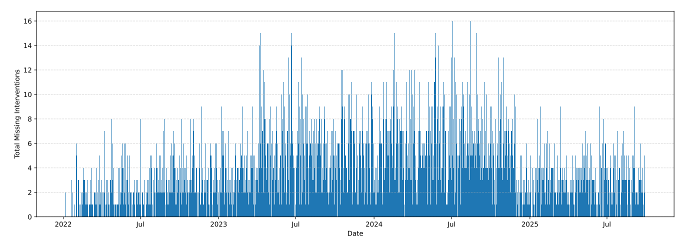

Fig. 10. Bar chart indicating the number of missing intervention time values in timeline

incident density in Bayraklı, compensating for its smaller number of streets relative to other districts.

Finally, Buca ranks second in the number of distinct streets but eighth in average incidents per street. A similar pattern appears in Balçova, Narlıdere, and Çiğli as well, indicating lower incident density despite many distinct streets. This makes these districts relatively less important when prioritizing preventive interventions.

Beyond street counts and incidents per street, these patterns may also reflect underlying exposure and built-environment differences between districts. For example, central districts such as Konak typically have higher population and employment density, more through-traffic, and a denser road network, all of which increase opportunities for crashes. Likewise, a district with relatively few streets but a high incidents-per-street value (such as Bayraklı) could indicate concentrated traffic on specific arterials or local design and control issues at key junctions, rather than simply a larger number of roads.

Figure 9 illustrates the relationship between the monthly number of distinct streets and total incidents, along with their ratio, computed by dividing the total number of incidents by the number of distinct streets each month. The top panel combines a bar chart of total monthly incidents and a bar chart of the monthly count of distinct streets, where both gradually decline across the period. The bottom panel plots the ratio of total incidents to distinct streets. This series remains fairly steady, typically between 10 and 16, which aligns with the gradual decrease trend shown in the top and middle panels. In summation, the decrease in total incidents appears to align with a decrease in the number of distinct streets involved.

Figure 10 shows how many entries in the intervention time column are missing each day. There are in total 5193 missing entries in the raw data set. On inspection, the figure shows entries for each day, indicating that missing entries are uniformly distributed over the timeline.

Figure 11 shows the distribution of the duration between accident time and intervention time. The figure follows a right-skewed distribution, indicating that incidents with short intervention times are more common than those with long intervention times.

After examining the patterns in the missing intervention time entries, it is observed that there are no time periods where data was missing in bar blocks. In conclusion, it is decided to impute these missing intervention time values. To achieve this goal, first of all, the median duration between accident time and intervention time is calculated within detailed groups defined by accident type, location, time interval, street and working day status. Then, missing values are imputed. In the Figure 12, the distribution of intervention time with imputed values can be seen.

Figure 13 plots incident occurrence time against intervention time in minutes. Blue points show points with intervention

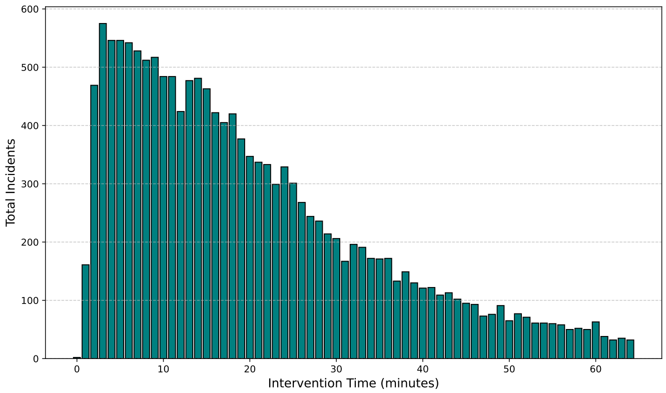

Fig. 11. Demonstrating total incidents in different intervention times

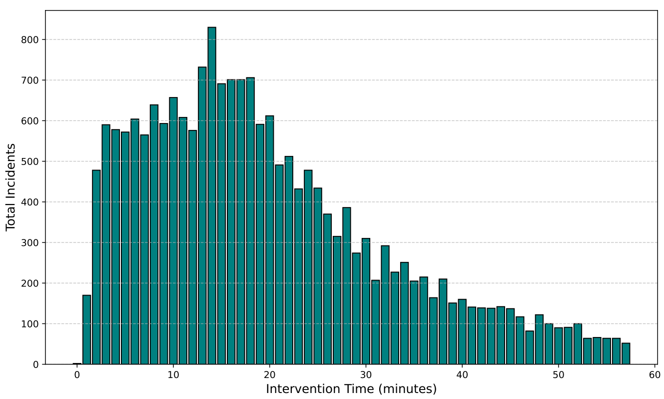

Fig. 12. Illustrating total incidents in different intervention times after imputing

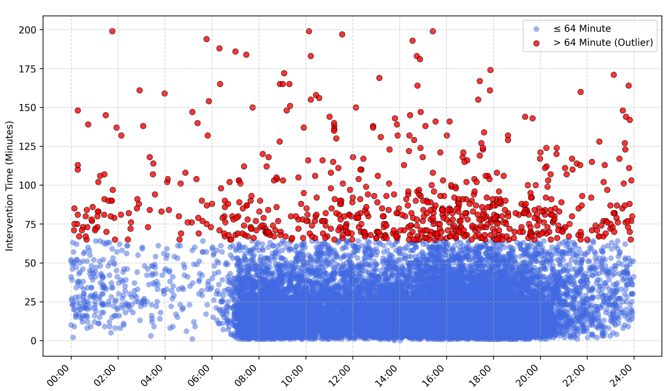

Fig. 13. Scatterplot Indicating Incident Time and Intervention Time Relations

times under 64 minutes and red points mark outliers with intervention times above 64 minutes. The 64 minute cutoff was derived from the intervention-time distribution using the standard boxplot rule: the interquartile range was 22 minutes, yielding an upper outlier threshold of 64 minutes and a lower bound of -24 minutes, which was truncated to 0 since negative values are not meaningful. In addition, 16 records with intervention times exceeding 200 minutes were treated as extreme anomalies and discarded from the analysis to prevent distortion of the computed statistics. The scatterplot suggests that during high-traffic hours the outlier rate is lower, whereas during late-night and early-morning periods which are outside of regular working hours, display a higher outlier proportion, which could be explained by lack of available staff that could concern incidents.

Figure 14 provides heatmap, demonstrating rate of correlation between selected attributes. To construct the heatmap, we first selected the rows with the frequent incident types: vehicle breakdown, injury/fatal accident and property damage only accident, then partitioned the timeline into intervals with approximately equal incident counts. This procedure resulted in six time intervals, each containing approximately 3,400 incidents. The first interval covers the late evening to early morning period (20:00–07:54). The second interval spans the early morning hours (07:55–10:32). The third interval corresponds to the late morning and early afternoon (10:33–14:23). The fourth interval represents the mid-afternoon period (14:24–16:30). The fifth interval includes the late afternoon and early evening (16:31–18:01). Finally, the sixth interval covers the early evening period (18:02–19:59).

Indicator columns were then created to distinguish between the most frequent incident categories: breakdown-related events, injury or fatality cases, and property-damage-only incidents. Next, normalization process is applied to achieve reasonable results. Because these features are nominal, we applied one-hot encoding to obtain numeric representations, followed by normalization to conduct healthy comparisons across variables. Subsequently, encoded variables are multiplied with their respective total incident numbers.

The heatmap reveals several interesting patterns. Injury or fatality-related incidents shows a relatively high positive correlation number of 0.47 with the earliest time interval. Since that interval comprises 20:00 to 07:54 hours, the injury incident rates are high as supported in Figure 7. In addition, property-damage-only events are more common during the mid-morning to late-afternoon period (approximately 07:55–16:30), with correlation values reaching up to 0.33. Finally, breakdown-related incidents display higher occurrence rates in the late-afternoon and evening hours (16:31–19:59), showing correlations of up to 0.28.

After discussing the figures, some additional findings about the attributes are discussed. In the GitHub repository of this project[2] and in Figure 14, there is a correlation matrix of

[2]https://github.com/HuseyinYontar/Analyzing_Traffic_Incidents_in_Izmir/blob/main/heatmap/onehot_heatmap.pdf

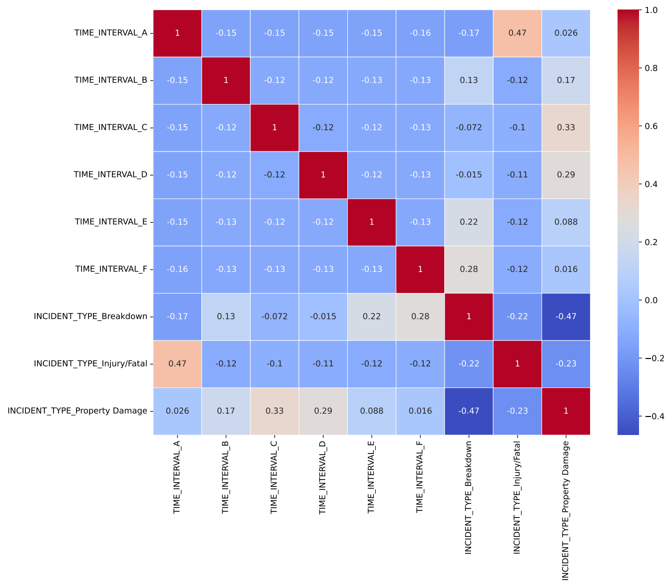

Fig. 14. Correlation heatmap of time intervals and several incident types

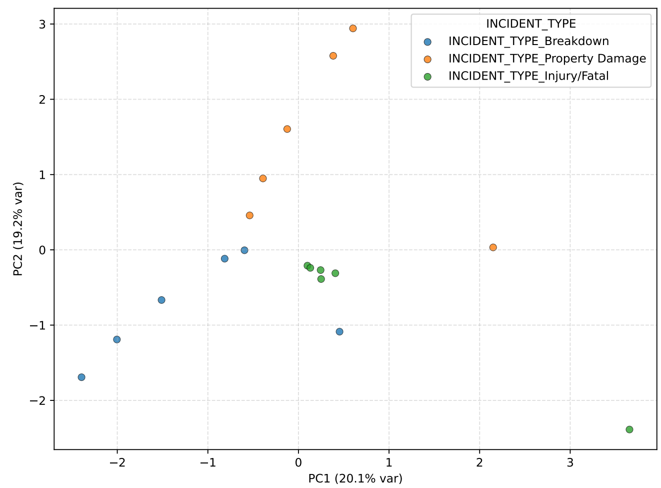

Fig. 15. Principal Component Analysis (PCA) between different incident types

some one-hot encoded categorical features. The purpose of this analysis is to identify potential relationships between different time intervals and incident types.

After searching for correlations between the attributes, in order to predict the type of the incidents, Random Forest Classifier and Decision Tree Classifier with Sequential Feature Selector were used for attribute selection. It is not ended with a significant result when the selection done with Random Forest Classifier. For the Decision Tree Classifier, the best result was founded in the columns: Street, Direction, District, Season, isHoliday and Time Interval. The optimal subset was reached after 27 features with a best cross-validation accuracy of 0.5181. The result can be founded in the GitHub repository[3].

In the project's GitHub repository[4], we conducted Principal Component Analysis (PCA). First, we applied data-cleaning procedures and removed records with missing values. Next, we performed one-hot encoding to convert categorical variables to numeric form. To examine alternative correlation structures, we ran PCA on both the original Excel dataset and an edited version that included derived attributes such as intervention time, day type, and season. The best resulting PCA yielded with the dataset used in the correlation matrix contains incident types that can be seen in Figure 15, from which a clear clustering of injury and fatal crashes can be observed.

V. Classification / Regression

For the classification task, the objective was to predict whether a future incident would be life-threatening or non-life-threatening. To achieve this goal, several supervised learning algorithms were employed, including Random Forest, k-Nearest Neighbors (KNN), Multilayer Perceptron (MLP) neural network, Logistic Regression, and Decision Tree. These methodologies demanded preparation of the dataset to make it suitable for classification tasks. Firstly, some types of recorded instances are removed to reach a simpler dataset that is clean from inconsistent or invalid incident type records. This goal was achieved by selecting four high-frequency categories in the raw data set and removing records that belong to other incident types: injury, fatal, property-damage-only, and vehicle breakdown. Then, injury and fatal crashes were merged into a single injury/fatal category, representing life-threatening incidents, while property-damage-only crashes and breakdowns were grouped as non-life-threatening events. As a result, the target variable is a binary indicator of incident criticality, enabling authorities to prioritize preventive measures according to the anticipated severity of future incidents.

In addition to the preprocessing applications, remaining records were augmented by using the incident timestamp. As a result of this step, the day of week, month, day type (weekday, weekend, or public holiday), and working day columns were added as new separate columns. Moreover, the incident hours were encoded using two complementary approaches. First, the incident time was discretized into hourly intervals, represented as [0-1), [1-2), . . . , [22-23), [23-24). Secondly, the interval-based discretization that was previously discussed in the heatmap analysis was reused as an additional attribute. Furthermore, seasonal information about the incidents was added. Finally, to capture spatial structure of the districts, street and location information were used to detect the district attribute manually.

In the classification analysis, only a small portion of the data set was used, consisting of 5554 incidents. The goal behind this choice was to mitigate class imbalance by applying a simple undersampling scheme. This method was motivated by the fact that life-threatening incidents consist of a small portion of the whole set. Accordingly, all life-threatening incidents

[3]https://github.com/HuseyinYontar/Analyzing_Traffic_Incidents_in_Izmir/blob/main/attribute_selection/sfs_decision_tree_accuracy_curve_without_cadde.pdf

[4]https://github.com/HuseyinYontar/Analyzing_Traffic_Incidents_in_Izmir/tree/main/pca

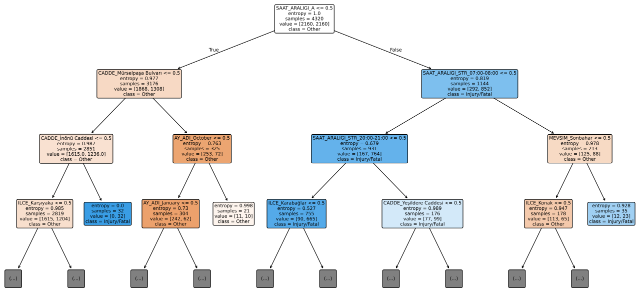

Fig. 16. Decision Tree model, indicating the severity of the incidents

were retained, and an equal number of non-life-threatening incidents were randomly sampled to form a balanced data set. The training portion of the data set was composed of 2160 life-threatening and non-life-threatening incidents which holds approximately 77% of the selected data. The remaining incidents were used in test set with 617 incidents per class, forming a group of 1234 observations in total which holds around 23% of the data set. After all, applying the undersampling procedure helped the model by reducing the risk of the classifiers being dominated by the majority class.

Table II shows the performance assessment of the classification models using overall accuracy, F1-score, class-wise precision and recall, as well as the confusion-matrix components (TN, FP, FN, TP).

Among the models, MLP classifier achieved the highest 0.7382 accuracy, with precision and recall values for both classes: (Life-threatening: precision 0.79, recall 0.65, F1-score 0.71; Non-life-threatening: precision 0.70, recall 0.82, F1-score 0.76). Logistic Regression obtained a comparable 0.7253 accuracy and followed by Random Forest with 0.7083 accuracy.

Another model trained in the classification analysis was a decision tree which is shown in Figure. When the decision tree model is examined, the most important attribute is found to be time interval, covering the late evening to early morning period (20:00–07:54). Full version of the decision tree can be founded in the Github repository[5].

In addition to the classification analysis, several regression methods were employed to predict monthly total incident counts using supervised learning. To prepare the data set for regression models, a univariate time series of daily incidents was created by applying a discretization method which involves grouping incidents by calendar dates and then counting the number of events per day. Daily counts were then aggregated to obtain monthly total incident counts, which were modelled as a continuous response variable using several supervised learning methods, including conventional regression algorithms and a neural-network model.

[5]https://github.com/HuseyinYontar/Analyzing_Traffic_Incidents_in_Izmir/blob/main/classification/decision_tree_for_different_types/decision_tree_full.pdf

To capture systematic differences between working days and holidays, a day-type feature was defined to indicate whether a given date is a weekday, weekend, or special day (public holiday), a seasonality feature indicating whether the observation belongs to winter, spring, summer, or autumn, and a month-name feature indicating the calendar month of the observation. These nominal variables were then transformed into a set of binary indicator features via one-hot encoding, allowing the models to exploit these patterns, which resulted in an improvement in the predictive accuracy of the regression models that is validated by the decrease in mean absolute percentage error.

The models were trained using the data starting from 23 December 2021 until the end of January 2025. The training period covers 1,136 days, which constitutes approximately 85% of the data. The remaining data between February and August 2025 were selected as the test set, covering 212 days, which corresponds to approximately 15% of the instances. This split was chosen so that the models were trained on a long historical period while being evaluated on a more recent interval with different seasonal patterns.

The regression analysis on this prepared data was conducted by using multiple models. Two of these were MLP Neural Network and Linear Regression. The third model was selected among different time series models: Auto-regressive (AR), Moving Average (MA), Exponential Smoothing, Auto-regressive Moving Average (ARMA), Auto-regressive Integrated Moving Average (ARIMA) and Seasonal Auto-regressive Moving Average (SARIMA); the model with the lowest MAPE was chosen as the final time-series model. All models were trained on the same training set.

During evaluation, all models were tested on the test set using a day-by-day prediction approach. For each day t in the test period, the input for the models was constructed by using the actual observations from the previous 21 days ($t - 21, \ldots, t - 1$), and a one-day-ahead forecast for day t was generated. This procedure was repeated sequentially for every day in the test set.

To obtain performance metrics, the prediction errors were then aggregated on a monthly basis. For each month in the test period, the Mean Absolute Percentage Error (MAPE) and Root Mean Squared Error (RMSE) were computed using the daily observed and predicted values within that month. These monthly error measures were subsequently used to compare the performance of the different which can be observed in Table III.

For visualizing the findings of the regression models, several figures were plotted. Figures 17–21 illustrate out-of-sample predictions for the period February–August 2025 at five high-priority locations: Konak district, Gaziemir district, Anadolu Street, Yeşildere Street, and Mürselpaşa Boulevard. In Konak (Figure 17), the MLP model achieves accurate predictions, yielding a test MAPE of 7.20% and RMSE of 15.86, while the SARIMA specification attains a slightly lower MAPE (6.60%) but higher RMSE (18.50); linear regression performs noticeably worse, with 10.71% MAPE and an RMSE of

TABLE II
DETAILED PERFORMANCE METRICS OF THE MODELS TRAINED

Model	Accuracy	Precision		Recall		F1-score		Confusion Matrix			
		Life-threatening	Non-life-threatening	Life-threatening	Non-life-threatening	Life-threatening	Non-life-threatening	TN	FP	FN	TP
Random Forest	0.7083	0.73	0.69	0.66	0.75	0.69	0.72	464	153	207	410
KNN	0.6515	0.71	0.62	0.52	0.78	0.60	0.69	484	133	297	320
MLP (NN)	0.7382	0.79	0.70	0.65	0.82	0.71	0.76	507	110	213	404
Logistic Regression	0.7253	0.78	0.69	0.63	0.82	0.70	0.75	506	111	228	389
Decision Tree	0.7050	0.75	0.67	0.61	0.80	0.67	0.73	495	122	242	375

TABLE III
SUMMARY OF MONTHLY MAPE AND RMSE VALUES OF MODELS FOR TESTED LOCATIONS (FEBRUARY 2025 – AUGUST 2025)

Location	Neural network (MLP)		Linear regression		Time-series model*	
	MAPE (%)	RMSE	MAPE (%)	RMSE	MAPE (%)	RMSE
Konak	7.20	15.86	10.71	24.28	6.60	18.50
Gaziemir	20.01	9.77	30.14	14.07	10.75	6.94
Yeşildere Caddesi	17.56	8.54	34.07	15.85	22.65	13.31
Mürselpaşa Bulvarı	6.19	3.68	18.82	10.19	14.35	11.23
Anadolu Caddesi	12.20	7.83	11.27	7.68	8.90	5.71

*Konak: SARIMA(1,1,1,7); Gaziemir: ARIMA(0,1,1); Yeşildere Street: SARIMA(1, 1, 1, 7); Mürselpaşa Boulevard: ARIMA(2, 1, 2); Anadolu Street: ARIMA(2, 1, 2).

24.28. In Gaziemir (Figure 18), the Auto-ARIMA time-series model provides the lowest errors (10.75% MAPE, RMSE 6.94), the MLP model yields moderate errors (20.01% MAPE, RMSE 9.77), and linear regression again produces the largest errors (30.14% MAPE, RMSE 14.07). On Anadolu Street (Figure 19), only the ARIMA model achieves the lowest error rates, outperforming the MLP and linear regression models in terms of accuracy (8.90% MAPE, RMSE 5.71, compared with 12.20% MAPE, RMSE 7.83 for the MLP and 11.27% MAPE, RMSE 7.68 for linear regression). For Yeşildere Street (Figure 20), error levels are higher for all models, with test MAPE scores ranging from 17.56% to 34.07% and RMSE values between 8.54 and 15.85, where the MLP again performs best and linear regression worst. On Mürselpaşa Boulevard (Figure 21), the MLP provides the best fit with 6.19% MAPE and RMSE of 3.68, followed by the ARIMA model (14.35% MAPE, RMSE 11.23), and lastly linear regression with 18.82% MAPE and RMSE of 10.19. These location-specific plots indicate that neural networks and selected time-series models perform better at predicting monthly incidents compared to linear regression models. Figure 22 and Figure 23 summarize the overall comparison of these models by showing the mean absolute percentage error and root mean squared error of the three approaches across all five locations. The bar charts reveal comparatively high MAPE and RMSE values for the linear regression model, supporting the prior findings. Yeşildere Street stands out as the most difficult location to predict, with all three models exhibiting relatively high MAPE and RMSE values, while Mürselpaşa Boulevard and Anadolu Street shows small error rates overall. These patterns indicate non-linear and time-series models are better suited than a simple linear specification, and that model choice should be tailored to individual locations rather than relying on a single global model.

Fig. 17. Comparison of observed and predicted monthly traffic incident counts in Konak between February 2025 and August 2025. The panels show (top) a multilayer perceptron neural network, (middle) a linear regression model, and (bottom) the best-performing SARIMA specification, with model predictions plotted against the observed monthly incident counts.

Fig. 18. Comparison of real and predicted monthly traffic incident numbers in Gaziemir between February 2025 and August 2025. The panels show (top) a multilayer perceptron neural network, (middle) a linear regression model, and (bottom) an ARIMA time series model specification, with model predictions plotted against the observed monthly incident counts.

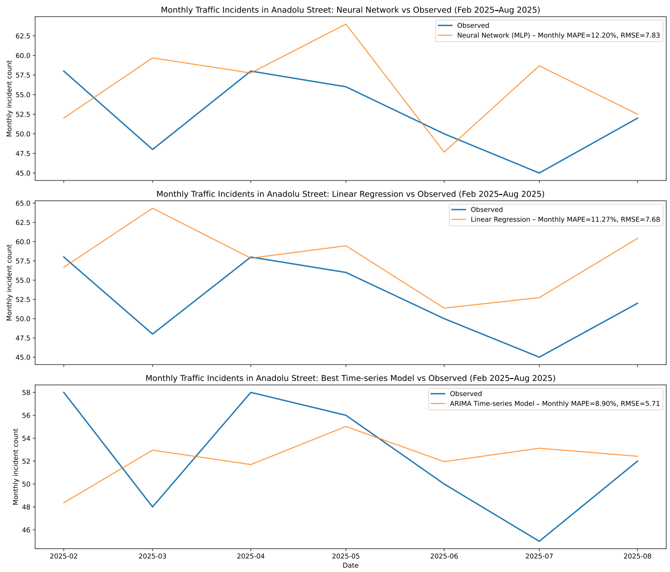

Fig. 19. Side-by-side comparison of actual and model-predicted monthly numbers of traffic incidents on Anadolu Street between February 2025 and August 2025. The panels display (top) a multilayer perceptron neural network, (middle) a linear regression model, and (bottom) an ARIMA model, with each model's predictions overlaid on the observed monthly incident counts.

Fig. 20. Monthly traffic incident totals in Yeşildere Street: observed values versus regression model estimates between February 2025 and August 2025. Shown in the panels are, from top to bottom, a multilayer perceptron neural network model, a linear regression model, and a SARIMA time series model, each with its monthly incident predictions plotted against the observed values.

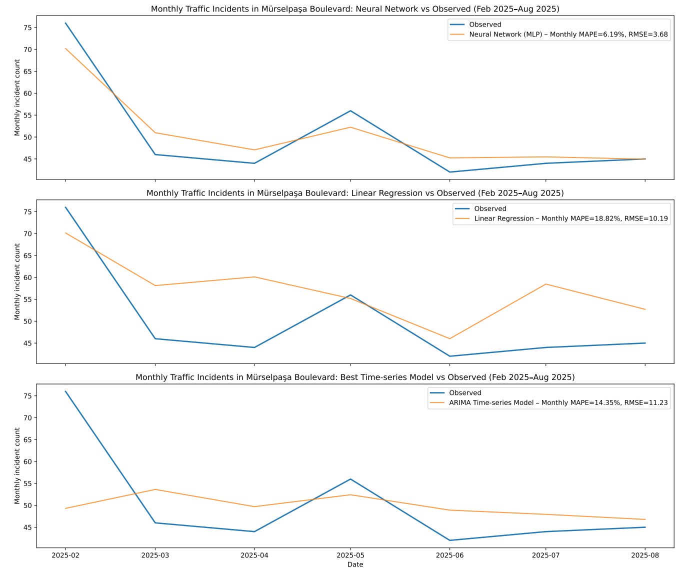

Fig. 21. Comparison between observed and forecasted monthly traffic incident counts on Mürselpaşa Boulevard between February 2025 and August 2025. In the panels, the top plot corresponds to a multilayer perceptron neural network, the middle to a linear regression model, and the bottom to an ARIMA time series model, with predicted monthly incidents plotted with on the observed data.

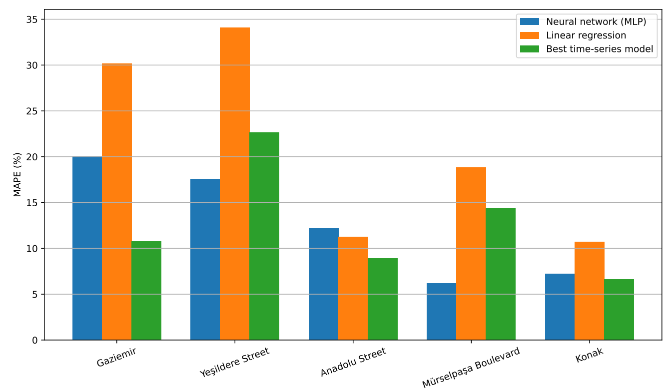

Fig. 22. Mean absolute percentage error (MAPE) of neural network, linear regression, and best time-series models for monthly incident predictions on selected streets and districts.

VI. Association Mining

Describe the association mining algorithms that you have applied to your data set Present and discuss your results

VII. Clustering

Describe the clustering algorithms that you have applied to your data set Present and discuss your results

VIII. Ensemble Learning

Describe the ensemble learning algorithms that you have applied to your data set Present and discuss your results

IX. Conclusion

Conclude your report by summarising your findings

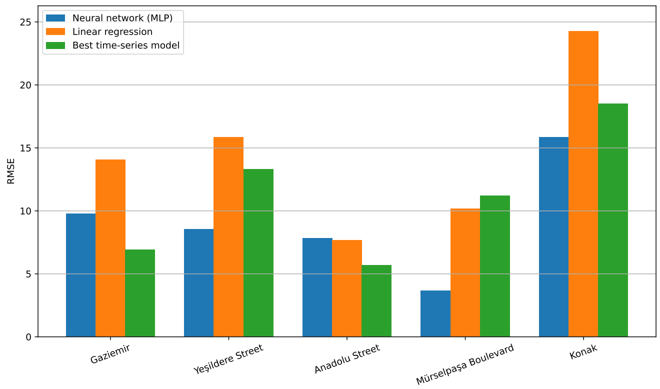

Fig. 23. Root mean square error (RMSE) of neural network, linear regression, and best time-series models for monthly incident predictions on selected streets and districts.

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